Disease Prevention, control and eradication

Goal: Improve knowledge and skills on disease prevention

Specific objectives:

- Explain the purposes and levels of disease prevention
- Explain the approaches and challenges to disease prevention

Prevention – is defined as intervention or interventions that interrupt the web of causality leading to one or more aspects of ill health.

Preventive healthcare – these are measures taken for disease prevention, as opposed to disease treatment. It is also called preventive medicine or prophylaxis. Each year, millions of people die preventable deaths.

Purpose of disease prevention

- Prevention is better than cure
- Stop new cases of disease from occurring
- Stop existing cases of diseases from spreading

Primary level prevention

- Here the goal is to protect healthy people from developing a disease or experiencing an injury in the first place. For example:
  - education about good nutrition, the importance of regular exercise, and the dangers of tobacco, alcohol and other drugs
  - education and legislation about proper seatbelt and helmet use
  - regular exams and screening tests to monitor risk factors for illness
  - immunization against infectious disease
  - controlling potential hazards at home and in the workplace

Secondary level prevention

- These interventions happen after an illness or serious risk factors have already been diagnosed.
- The goal is to halt or slow the progress of disease (if possible) in its earliest stages; in the case of injury, goals include limiting long-term disability and preventing re-injury. For example:
  - telling people to take daily, low-dose aspirin to prevent a first or second heart attack or stroke
  - recommending regular exams and screening tests in people with known risk factors for illness
  - providing suitably modified work for injured workers

Tertiary level prevention

- This focuses on helping people manage complicated, long-term health problems such as diabetes, heart disease, cancer and chronic musculoskeletal pain.
- The goals include preventing further physical deterioration and maximizing quality of life. For example:
  - cardiac or stroke rehabilitation programs
  - chronic pain management programs
  - patient support groups

Preventive techniques

- Prevention is the first line of defense against disease.
- At least 8 preventive techniques are available for disease prevention:
  - ✓ Exclusion - of causative agents of disease from specific geographic areas, or quarantine.
  - ✓ Control methods - such as immunization, environmental control, and chemical agents to protect specific animal populations from endemic diseases, diseases normally present in an area.
  - ✓ The mass education - of people about disease prevention.
  - ✓ Early diagnosis - of illness among members of an animal population is important so that disease manifestations do not become too severe and so that affected animals can be more easily managed and treated.
  - ✓ Recognition - of the relationship among high risk population
  - ✓ Health protection measures - and the second group is the health promotion measures. The two approaches are complementary and interactive, and it is unlikely that one would be effective without the other. A third group of measures include preventive health services.
  - ✓ Health promotion measures - include activities that individuals and communities can use to promote healthy life style. For example: public education programs, health warning labels, advertising and marketing, and media portrayals.

Simple things you can do to prevent illness.

- o Keep Immunizations up to date
- o Wash your hands often
- o Be aware of what you eat, and be careful how you prepare it
- o Use antibiotics exactly as prescribed
- o Report to your doctor any worsening infection that does not get better after you take a prescribed antibiotic
- o Be cautious around wild and domestic animals that are not familiar to you
- o Avoid areas of insect infestation
- o Avoid unsafe unprotected sex and injection drug use
- o Stay alert to disease threats when you travel
- o When sick, allow yourself time to heal and recover.

The strategic directions to disease prevention

- The strategic directions provide a strong foundation for all of our nation's prevention efforts and include core recommendations necessary to build a prevention-based strategy.
- Each Strategic Direction guides actions that will demonstrably improve health. Together, the Strategic Directions create the web needed to fully support in leading longer and healthier lives.
- The four Strategic Directions are:
  - o Healthy and Safe Community Environments
  - o Clinical and Community Preventive Services
  - o Empowered People
  - o Elimination of Health Disparities

- Within the Strategy, the priorities provide evidenced-based recommendations that are most likely to reduce the burden of the leading causes of preventable death and major illness.
- The Priorities are designed to improve health and wellness of the population, including those groups disproportionately affected by disease and injury.
- The seven Priorities are:
    - Tobacco Free Living
    - Preventing Drug Abuse and Excessive Alcohol Use
    - Healthy Eating
    - Active Living
    - Injury and Violence Free Living
    - Reproductive and Sexual Health
    - Mental and Emotional Well-Being

A Plan for Better Health and Wellness

The overarching goal of the National Prevention Strategy is to increase the number of people who are healthy at every stage of life. The strategy provides evidence-based recommendations that are fundamental to improving the nation's health through the active engagement of all sectors of society to help achieve four broad strategic directions:

- Building Healthy and Safe Community Environments - Prevention of disease starts in our communities and at home; not just in the doctor's office. For example, businesses and employers can adopt practices to encourage their workforce to increase physical activity and reduce pollution (e.g., workplace flexibility, rideshare and vanpool programs, park-and-ride incentives, travel demand management initiatives, and telecommuting options).

- Expanding Quality Preventive Services in Both Clinical and Community Settings - When people receive preventive care, such as immunizations and cancer screenings, they have better health and lower health care costs. For example, expanding the linkages between clinical and community prevention efforts, such as diabetes prevention programs that support preventive efforts among underserved groups and can improve access to preventive services.

- Empowering People to Make Healthy Choices - Policies and programs can make healthy options the easy and affordable choice, and when people have access to actionable and easy-to-understand information and resources, they are empowered to make healthier choices. For example, health care professionals can use multiple communication tools (e.g., mobile phone applications, personal health records, and credible health websites) and culturally competent methods to support more traditional written and oral communication.

- Eliminating Health Disparities - By eliminating disparities in achieving and maintaining health, we can help improve quality of life for all Americans. For example, health care providers can train and hire more qualified staff from underrepresented racial and ethnic minority groups and people with disabilities.

Role of environment and individual approaches to disease prevention

- *Ecosystem* – Environment is where man, pathogens (virus, bacteria, fungi and protozoa) live. The interaction between man with environment enable transfer of germs to human

system

- *Difficulties in changing behavior*- To prevent disease we increasingly ask people to begin do things that they have not done previously, to stop doing things that they have been doing for years and to do more of some things and less of others

- *Pattern of disease rate* - Some groups often have a characteristic over time even though individuals come and go from these groups. If groups have different rates over time, there may be something promotes or discourages disease among individuals in those groups

- *Socio-economic status*- A consistent finding dated from the twelfth century, is that people in the lowest socioeconomic groups have the highest rates of morbidity and mortality, whether the socioeconomic status was studied in relation to education, income, or occupation: the lower the level the higher the death and morbidity rate.

- Marital status - It has been known for many years that people who are not married- whether single, separated, widowed or divorced- have higher mortality rates than married people.

- Gender - One of the most well-established facts among students of health and disease is that men have higher mortality rates than women. This excess of male deaths occurs at every age and for every major cause for which comparison is possible.

Health education

- Is any is any combination of learning experiences designed to help individuals and communities improve their health, by increasing their knowledge or influencing their attitudes.
- A level of education plays a major role in health education. Different levels of education has different levels of perception and understanding
- Target audiences are those that the education is intended for. They can be:
    - Primary target audience – people who are directly affected
    - Secondary target audience – people who are indirectly affected
    - Tertiary target audience – people who are indirectly affected and can influence policy
- Some of the education outlets in the community include chief's baraza, churches, schools, social places, boda-boda terminals etc.
- Promoting health and preventing disease in the population and in the wider community involves the following:
    - *Demonstrate effective, patient-centered counseling*
        - ✓ Interventions that address patients' personal health behaviors and practices are vitally important.
        - ✓ Clinicians can help individuals change their health-related behaviors by using efficient, patient-centered counseling strategies that are based on an understanding of how people change.

    - *Understand the principles of and procedures for screening.*
        - ✓ Another central health promotion/disease prevention strategy that students need to be skilled in is screening, or testing for health problems in the absence

of symptoms or signs indicating the presence of such a problem.
    - ✓ Students need to learn (1) when screening is appropriate, (2) what procedures to use and how to use them effectively, and (3) how to interpret and use the results of screening tests.

- *Describe the importance and use of immunization and prophylaxis.*
    - ✓ Understand the important role that widespread childhood immunization has played in drastically reducing the incidence of many illnesses, including diphtheria, pertussis, tetanus, poliomyelitis, measles, mumps, and rubella.
    - ✓ In addition, learn which childhood immunizations are essential, when and how to administer them, and potential risks.
    - ✓ Knowledge of the appropriate use of adult immunizations is also necessary.

- *Implement strategies to promote the health of the population.*
    - ✓ Although health care professionals traditionally have focused on individual patients who present themselves for care one at a time in the office or clinic, students preparing for practice in today's health care system also must learn to focus on the care of populations of patients.

- *Engage in public education and advocacy to prevent disease.*
    - ✓ For some health problems, preventive interventions aimed at the wider community may be especially appropriate and effective.
    - ✓ Such problems include unintended pregnancy in adolescents, youth violence, and domestic violence, initiation of tobacco use, sexually transmitted diseases, healthy dietary practices, and unintentional injuries.

The challenges to disease prevention

Political challenges

- The scope for the prevention of diseases is enormous.
- There will be scope for prevention of all types and at all levels, and much of this will require political change.
- Smoking-related diseases are increasing in developing countries where tobacco companies have been promoting sales.
- Widespread disease and death still occur in many countries which don't have a clean and plentiful water supply.
- Wars and armed conflict have hindered or destroyed the development of basic infrastructure in many parts of the world. Worldwide, it is evident that measures at government level will yield the greatest gains.

Specific challenges

- Many of these are clinical or individual challenges.
- This may include the barriers to prevention. Articles of particular note include:

Taxation and cost issues

- Consumption may fall so much that total revenue from that source falls. This can be

politically challenging with industry concerned about economic damage.

- Most interventions have costs and do not have the advantages of revenue generation associated with, for example, taxation on tobacco. This includes screening programmes or advertising campaigns.
- Immunization programmes cost money to set up in private health institutions

Demographic issues

- A growing elderly population puts more strain on health and social services. A significant number of people now spend a quarter of their lives as pensioners retired from working life.

Personal barriers

- Cost and quality
- Information
- Exercise
- Social, educational and cultural barriers

Administrative barriers